Name: **Student Number:**

MA2071: Multivariable Calculus

Midterm Exam - 100 marks - Show all work

1) (i) State (but do NOT derive) the equation of a plane P through the point (x_0, y_0, z_0) with normal vector $\mathbf{n} = \langle a, b, c \rangle$.

(ii) True or False: two planes are parallel if their normal vectors are parallel.

(iii) Find the distance between the planes $10x + 2y - 2z = 5$ and $5x + y - z = 1$ (Hint: are these planes parallel?).

2) (i) Either compute the following limit or prove that it does not exist:

$$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2 + y^4}.$$

(ii) Find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$ if $x^3 + y^3 + z^3 + 6xyz = 1$.

3) (i) Find all points at which the direction of fastest change of $f(x, y) = x^2 + y^2 - 2x - 4y$ is $\mathbf{i} + \mathbf{j}$.

(ii) Find the equation of the tangent plane to the surface $yz = \ln(x + z)$ at the point $(0, 0, 1)$.

4) Find the volume of the tetrahedron bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$ and $z = 0$.